

TECHNOLOGY BRIEF · COMPANY 2 OF 2

Isaacus

Foundational legal AI models — embeddings, zero-shot clause classification, reranking, and graph enrichment — and how they slot into DealDiligence's retrieval and extraction layers.

COMPANY

Isaacus (isaacus.com)

CATEGORY

Legal embedding & classifier models

ACCESS

API · AWS · self-hosted

Evaluation & integration options

Already named in our ARCHITECTURE.md

EXECUTIVE SUMMARY

What this is, and why it matters to us

Isaacus is an Australian foundational legal-AI company that builds small, fast, **specialized legal models** — not chatbots. Their **Kanon** family covers the exact infrastructure layers DealDiligence needs: turning contracts into searchable vectors, detecting clauses with zero examples, reranking results, and extracting structured graphs. Their legal-tuned embedder ranks **#1 on the industry's main legal benchmark**, beating OpenAI, Google, and Voyage while being faster and smaller — at **\$0.35 per million tokens**.

#1

On MLEB, the Massive Legal Embedding Benchmark — ahead of 20 models

+9%

More accurate than OpenAI's top embedder, >30% faster, on legal text

38

Jurisdictions of laws, cases & contracts in their training data

WHY IT'S DIRECTLY RELEVANT — IT'S ALREADY IN OUR PLAN

Our own `ARCHITECTURE.md` names **Isaacus** as a candidate retrieval provider behind the `Retriever` interface (alongside Voyage / OpenAI). Today we run a "stuff-everything" retriever with no real semantic search. Isaacus is the drop-in that turns that into best-in-class *legal* retrieval — plus three more models that map onto clause detection, the knowledge graph, and verification.

Bottom line up front: where Company 1 (Stateful Swarms) is an *orchestration brain*, Isaacus is **battle-tested specialized infrastructure** — used by Microsoft, IBM, and the World Bank, serving 40B tokens/month, with SOC/ISO/IRAP compliance and self-hosting. The fastest win is swapping the embedder into our planned vector retriever; the highest-value win is using their zero-shot classifier to detect CUAD clauses and playbook red-flags cheaply, without an LLM call.

01 • THE COMPANY

Who Isaacus is

A foundational legal-AI startup with a deliberate thesis: **specialized models beat general-purpose chatbots** for legal work. They build focused, best-in-class tools — "effective, efficient, scalable" — rather than another GPT wrapper.

Traction & customers

- Used by **Microsoft, IBM, UC Berkeley, the World Bank**, and major universities
- **40 billion tokens** served monthly
- **4M monthly downloads** of their open-source tooling

Deployment & compliance

- Cloud API, **AWS Marketplace** (air-gapped SageMaker), and **self-hosted / on-prem**
- **SOC, ISO, and IRAP** compliance certifications
- Models small enough to run privately — even on-device

COMPLIANCE FIT

Self-hosted + SOC/ISO/IRAP maps cleanly onto our prod posture — Claude via Bedrock *inside the customer's AWS account*. Isaacus can run in the same isolated environment, so contracts never leave the client's perimeter. That's a strong story for regulated M&A buyers.

They also set the benchmarks

Isaacus authored the open-source standards the field is measured against — credibility that they build for retrieval quality, not hype:

- **MLEB (Massive Legal Embedding Benchmark)** — the largest legal-retrieval benchmark: 10 datasets across 6 jurisdictions (US, UK, EU, Australia, Singapore, Ireland) and 5 domains (cases, statutes, regulations, contracts, academia).
- **Legal RAG Bench** — evaluates retrieval-augmented generation on legal questions.
- **semchunk** — their open-source semantic chunking library, ~4M downloads/month, used across major tech companies (free for us to use in any embedding pipeline).

02 • THE PRODUCT FAMILY

The Kanon models

Five capabilities, each a small specialized model, billed per input token. Every one maps to a layer we already have or plan to build.

MODEL	WHAT IT DOES	\$ / 1M TOKENS
Kanon 2 Embedder	Legal embeddings for semantic search & clustering — #1 on MLEB	\$0.35
Kanon 2 Reranker	Re-orders retrieved passages by legal relevance to a query	\$0.35
Kanon Universal Classifier	Zero-shot clause/concept classification from plain-English criteria	\$1.00
Kanon Answer Extractor	Extractive QA — pulls exact verbatim answers from documents	\$1.50
Kanon 2 Enricher	"Graphitization" — documents → hierarchical knowledge graphs, sub-second	\$3.50

THE STANDOUT — ZERO-SHOT LEGAL CLASSIFICATION

The **Universal Classifier** needs no training examples. You write a plain-English statement — e.g. *"This clause entitles one to terminate an agreement in the event of circumstances beyond their reasonable control"* — and it scores how strongly each passage matches, with "startlingly accurate confidence scores." It can even assess *"whether an agreement unilaterally benefits one party"* — which is essentially automated red-flag detection. It comes in two sizes: **317M** params (+6% vs competitors) and a **136M** "Mini" (+12%, 441MB — small enough to run on a phone).

All Kanon models are built on a legal foundation model trained on millions of laws, regulations, cases, contracts, and papers across 38 jurisdictions — which is why they beat general-purpose models on legal text despite being far smaller. A proprietary global legal corpus, **Blackstone Graph**, is in development.

03 · THE FIT

How it maps onto DealDiligence

Unlike a brand-new architecture, Isaacus slots into components we already have. Each model upgrades a specific layer.

DEALDILIGENCE LAYER	ISAACUS MODEL	WHAT CHANGES
stuffRetriever (no embeddings, returns everything)	Kanon 2 Embedder	The named "PgVectorRetriever" swap — real legal semantic search
CUAD clause detection (Opus extraction)	Universal Classifier	Detect/verify clauses zero-shot, no LLM, \$1/M
Playbook red-flag matching	Universal Classifier	Score "does this unfairly favor one side?" deterministically
Haiku document-type classifier	Universal Classifier	Cheaper, legal-tuned doc categorization
Chat & task-attachment retrieval	Kanon 2 Reranker	Rank fact sheets/clauses before they reach Claude
Reconciliation → MasterEntity / Relationship graph	Kanon 2 Enricher	Sub-second graphitization of entities & relationships
Verify layer (citation validation)	Kanon Answer Extractor	Pull exact grounded quotes to check citations

THE CLEANEST ENTRY POINT

Our Retriever interface was *designed* for this swap — "when scale demands, swap in PgVectorRetriever or a provider (Voyage / Isaacus / OpenAI) behind the same interface; nothing else changes." Dropping in the Kanon 2 Embedder is a contained change with an immediate quality lift on chat and retrieval, and no rework anywhere else.

04 • WHAT WE COULD ACTUALLY DO

Potential implementations

Ranked by effort-to-value. Each carries an effort and impact rating.

1

Kanon 2 Embedder as our vector retriever

Effort: Low

Impact: High

Implement the planned `PgVectorRetriever` using Kanon 2 Embedder behind our existing `Retriever` interface, with their open-source **semchunk** for chunking. Immediately upgrades VDR chat and task-attachment retrieval from "load everything in scope" to true legal semantic search — better answers, fewer tokens into Claude. Lowest-risk, highest-certainty win.

2

Universal Classifier for clause & red-flag detection

Effort: Med

Impact: High

Use zero-shot classification to (a) **verify** the clauses Opus claims to have found — a cheap legal cross-check that strengthens our verify layer, (b) detect **playbook red-flags** by scoring plain-English statements against each clause, and (c) pre-screen long documents for clause presence before spending Opus tokens. Deterministic, legal-tuned, \$1/M — a strong complement to (not replacement for) Claude's structured extraction.

3

Kanon 2 Reranker on retrieval results

Effort: Low

Impact: Med

Add a rerank pass over retrieved fact sheets/clauses so the most relevant context reaches Claude first. Pairs naturally with implementation #1, cuts wasted context tokens, and sharpens chat answers — at \$0.35/M, effectively free at our volume.

4

Kanon 2 Enricher to accelerate the knowledge graph

Effort: Med

Impact: Med-High

Our reconciliation builds a `MasterEntity` + relationship graph via a Sonnet pass. The Enricher graphitizes documents into hierarchical entity/relationship structures in sub-second time — it could seed or accelerate that graph (and the Cytoscape explorer) more cheaply and consistently than a full LLM reconciliation each rebuild.

5

Self-hosted deployment for regulated clients

Effort: Med

Impact: Strategic

For SOC 2 / data-residency-sensitive buyers, run Kanon models air-gapped in the customer's AWS alongside our Bedrock deployment. Keeps the entire retrieval + classification stack inside the client perimeter — a differentiator in enterprise M&A sales.

05 • THE TWO COMPANIES TOGETHER

Isaacus vs Stateful Swarms — and why both

The two vendors operate at different layers of the stack. They aren't competitors for our attention — they're complementary.

DIMENSION	STATEFUL SWARMS (IQIDIS)	ISAACUS
What it is	Multi-agent orchestration "brain"	Specialized legal infra models
Layer	Coordination & reasoning	Retrieval, classification, graph
Maturity	Early — "Phase 0, unproven"	Production — enterprise customers
Access	MIT open source	Commercial API / self-host
Cost model	You run the models (any provider)	\$0.35–\$3.50 / M tokens
Best first move	MCP pilot on 2–3 deals	Swap embedder into retriever
Primary risk	Immature, uncalibrated	Vendor dependency, jurisdiction coverage

HOW THEY'D COMBINE

Isaacus supplies the **primitives** — legal embeddings, zero-shot clause scoring, graph enrichment — that any pipeline (ours today, or a swarm tomorrow) consumes. A future stateful-swarm reconciliation could use the Kanon Embedder for retrieval and the Universal Classifier as a cheap, deterministic "debt sensor" for clause presence. **Adopt Isaacus for the building blocks now; treat the swarm as an architecture to grow into.**

06 • THE HONEST ASSESSMENT

Risks & recommendation

Risks & caveats

- **Vendor dependency.** Embeddings are low lock-in (behind our `Retriever` interface, swappable) — but the Classifier and Enricher are more bespoke; build abstractions so we're not trapped.
- **Jurisdiction coverage.** Training spans 38 jurisdictions and MLEB covers US/UK/EU/AU/SG/IE — confirm strong coverage for the specific markets our deals concentrate in before relying on it.
- **It scores, it doesn't draft.** The Universal Classifier returns presence/confidence, not full structured extraction — it complements Claude's generative extraction, not replaces it.
- **Infra addition.** Embeddings require a vector store (`pgvector`) we don't run yet — modest new infrastructure, already on our roadmap.
- **Per-token cost at scale.** Cheap per call, but re-embedding large document sets adds up; like extraction, it's a one-time-per-document cost and should be cached/idempotent.
- **Smaller vendor.** An Australian startup vs a hyperscaler — though strong enterprise adoption, compliance certs, and self-hosting de-risk this materially.

RECOMMENDATION

Isaacus is the **lower-risk, faster-payback** of the two companies and is already earmarked in our architecture. **Move on implementation #1 (Kanon 2 Embedder as our vector retriever) as a near-term project** — it delivers a concrete chat/retrieval quality lift with contained scope. Trial the Universal Classifier in parallel as a verify-layer and red-flag enhancement. Keep self-hosting in our back pocket for the first regulated enterprise deal.

Combined verdict. Adopt **Isaacus** now for retrieval + classification building blocks; **Universal Classifier** as the longer-horizon orchestration upgrade. The two are complementary layers of the same future stack.

Sources: isaacus.com, docs.isaacus.com (capabilities, models, pricing), the Kanon 2 Embedder & MLEB announcements, and our own ARCHITECTURE.md retrieval note. Figures are vendor-reported; validate on our data during a pilot.